

Jacob Berger

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EDUCATION

Oregon State University

Ph.D. & M.S. in Artificial Intelligence

Honors B.S. in Applied Computer Science & Computational Mathematics

Corvallis, OR

Expected June 2029

Sept 2022 – June 2026

- **Honors:** Summa Cum Laude (both degrees) | **GPA:** 3.9+
- **Status:** Admitted to Accelerated Master's Program
- **Relevant Coursework:** Analysis of Algorithms, Machine Learning, Linear Algebra II. **Graduate:** Deep Learning, Operating Systems II, Numerical ODEs, Math of Data Science, Computational Number Theory

RESEARCH

Honors Thesis: Semantic Constraints in NLP

Researcher (Advisor: Dr. Lizhong Chen, STAR Lab)

Corvallis, OR

June 2025 – Present

- Led a first-author submission to COLM ("Parsimonious Decoding"), developing novel metrics to enforce rigorous semantic constraints in probabilistic language models to improve vocabulary legality.

Senior Capstone Project: AI for Home Automation

Project Team Member

Corvallis, OR

September 2025 – June 2026

- Collaborating to integrate Large Language Models with the Home Assistant IoT platform, enabling complex automations via natural language; selected for the OSU Engineering Research Expo.

EXPERIENCE

NLP SaaS Startup

AI Research Engineer

Portland, OR

March 2024 – Present

- **Grant Success:** Co-authored and secured a Phase I SBIR grant (Federal Research Funding) for R&D.
- Architected end-to-end data pipelines for training and fine-tuning specialized large language models.
- Engineered and deployed cloud infrastructure on AWS, GCP, and Vercel to train and host services.

MU Lanes and Games

Head Mechanic

Corvallis, OR

December 2022 – Present

- Managed maintenance projects and diagnostics for complex electromechanical systems.

MU Student Activities and Advisory Board

Vice-Chairman

Corvallis, OR

June 2024 – June 2025

- Oversaw budget preparation and facilitated strategic board meetings for OSU's Memorial Union.

SKILLS

- **Languages:** Python, C++, Java, Javascript, Matlab, Bash/Shell, x86 Assembly
- **Machine Learning:** PyTorch, HuggingFace Transformers, LLM Fine-tuning (LoRA), RAG Pipelines, RL
- **Systems & Infrastructure:** Linux (Debian), Proxmox VE, Virtualization (LXC/KVM), GPU Passthrough
- **Cloud & DevOps:** AWS (EC2/S3), Google Cloud Platform (GCP), Git, Vercel
- **Languages (Human):** Fluent in Spanish (9 years of language immersion education)

AWARDS

- **Putnam Math Competition Award:** Top 3 Finisher at Oregon State University.
- **Finley Academic Excellence Scholarship:** \$20,000 scholarship recipient for academic merit at OSU.

EXTRACURRICULARS

- **Competitive Math:** Competitor in OSU Putnam and COMAP Math Modeling competitions (implemented Cellular Automata to analyze spatial dynamics and ecosystem resilience).
- **Competitive Sports:** Founded OSU's national collegiate bowling team (All-Star qualifier); served as Rose City Rowing Club team leader (2-time national qualifier).